17 p and q are two numbers such that $p > q$

When you subtract 5 from p and subtract 5 from q the answers are in the ratio 5 : 1
When you add 20 to p and add 20 to q the answers are in the ratio 5 : 2

Find the ratio $p : q$
Give your answer in its simplest form.

$(p - 5) : (q - 5) = 5 : 1$

$(p + 20) : (q + 20) = 5 : 2$

1 mark
Express both ratios algebraically

$p - 5 = 5(q - 5)$, so $p - 5q = -20$ (1)

$2(p + 20) = 5(q + 20)$, so $2p - 5q = 60$ (2)

1 mark
Form equations from the ratios

Subtract (1) from (2): the $-5q$ terms cancel.

$(2p - 5q) - (p - 5q) = 60 - (-20)$

$p = 80$

1 mark
Eliminate q to find p

Substitute $p = 80$ into (1):

$$\begin{array}{rcl} 80 - 5q & = & -20 \\ \text{(-80)} & & \text{(-80)} \\ -5q & = & -100 \\ \text{(÷(-5))} & & \text{(÷(-5))} \\ q & = & 20 \end{array}$$

1 mark
Substitute to find q

$p : q = 80 : 20 = 4 : 1$

Check: $75 : 15 = 5 : 1$; $100 : 40 = 5 : 2$.

1 mark
Simplify the required ratio

4 : 1

(Total for Question 17 is 5 marks)

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